

Layoffs Across the World From 2020-2026

Purpose:

This was analysis done on the dataset “[Layoffs Dataset](#)” downloaded from [Kaggle.com](#). The purpose of this analysis was to learn and familiarize myself with the basic tools and processes that a data analyst might utilize in the day to day operations.

The Question:

How did layoffs affect different industries and countries across the world?

Key Findings:

Findings are presented in two parts, a raw overview of the entire data set, and a Controlled Analysis where we control for company shutdowns and year/industry/country aggregations are filtered out where the number of reported companies are less than 5.

Raw Overview:

1. Of the companies who had identified industries from 2020 through 2026, Retail, Hardware, and the Consumer industries all had the highest numbers of layoffs at 106.7k, 106.2k, and 97k with 31.53%, 20.78%, and 29.62% respectively.
2. In terms of the highest percentage of current workforce laid off, Aerospace, averaging 45.40% is more than 6 points ahead of AI in second place with 39.35%. In third we have the food industry with 38.97%.
3. If we exclude shutdowns, from 2020 through 2026 the United States tops the list for countries with the most number of layoffs with 443k, dwarfing the rest of the list combined. India takes second with about 30.7k, and third place is the Netherlands with 17.8k.
4. When we order by average percent of layoffs per country from 2020 through 2026, once again excluding shutdowns, we find Mexico in first with an average of 29.33% with 170 total layoffs, Nigeria in second with 27% with 1.6k, and India in 3rd with 23.77% and 31k. But Nigeria and Mexico's low total layoffs suggest these percentages may not be representative.

5. Looking at shut down companies per industry per year, finance in 2024 had the most at 13 shutdowns, healthcare in 2023 takes second with 12 shutdowns, and there is a tie for third with Finance 2023 and Food 2023 at 11.

Controlled Analysis:

6. Narrowing down to how each industry who had more than 5 companies reported fared per year, excluding companies who shut down, Hardware in 2024 and 2023 top the charts for the most total layoffs with 25k and 21k, suggesting a sustained multi-year decline rather than a single catastrophic event with Intel and Dell having a combined 28k over those 2 years, followed by Consumer in 2022 with 19k.
7. When looking at percent laid off for industries who had over 5 companies report that year, excluding companies who shut down, 2020 is the worst year, accounting for the top 4. Fitness with 33.29%, Real Estate with 32.92%, Food with 31.31%, then Consumer with 31.05%. This is consistent with a broad, industry-wide disruption.
8. Looking at the same criteria as the Key Finding 4 in the Raw Overview section, but also adding the criteria of needing to have substantial layoffs, above 10k total, we find that India then tops the list having 23.77% and a total of 30.7k, in second is the United Kingdom with 22.48% and 14.5k total, and Israel in third with 21.09% and 11.8k total.
9. Finally, looking at shutdowns for every country, we have the United States in first with 191 companies, India in second with 42, and the United Kingdom in third with 16.

Methodology:

The dataset was downloaded from Kaggle as a CSV file. It was then duplicated and connected to a python script called "Data Cleaning" where I used the python library called pandas to clean the data, the script can be found in the GitHub repository. After the data was cleaned in python and saved to a new CSV file, that file was uploaded to my Azure SQL Server using the SQL Server extension on VS Code. The data was then analyzed by querying that database through VS Code.

The decision was made to set a lower bound criteria for some of the metrics since outliers could massively swing the data and mislead the reader, so for most of the Controlled Analysis key findings we limit whatever we are aggregating by (industry/year/country) to have more than 5 reporting companies. I also excluded the 'Other' industry label from all queries instead of picking and choosing when I do and don't to keep the analysis consistent. I also separated when I was analyzing shutdowns and I wasn't, and this is because a few shutdowns can not only massively skew the results, but also because a company shutting down is not the same event as a layoff, and thus should be analyzed as such. Finally, for key finding 8, the decision was made to pick countries with "substantial layoffs" as in key finding 7 Mexico and Nigeria both had miniscule total layoffs but a high percent of layoffs, so I decided to do key finding 8 to clarify and contextualize key finding 7.

Limitations:

The first and most obvious limitation is the data is only limited to companies that were recorded in the dataset. For example, the US dominated the list of total layoffs but over an order of magnitude, but that's also because it had the most recorded companies, but without knowing the total number of companies in every country, we truly can't know how significant. The analysis also relies on reported data, if it is not reported by the sources used in the dataset, then I can't analyze that. We also had the issue of incomplete data, even after dropping all of the companies that had NULLs or no values for both the total layoffs and percent layoffs, which ended up being around 724 companies, we still had 810 companies with missing total laid off values, and 932 with missing percent layoffs. We decided to keep the companies where only one was missing, but whatever column of data was missing is preventing that company from being included completely in the analysis. I also decided to limit analysis of industry and country layoffs where the number of companies was above 5, and this was done because one or two companies that had a high number of total layoffs and/or percent laid off then could, and did, skew the results and were more indicative of outliers than a general trend.

Main Takeaways:

Layoffs were widespread, with no obvious through-line pointing to any specific country or industry. While the US dominates in raw numbers, countries like Israel and Nigeria lead on percentages. Depending on what metrics and filters you apply, you can draw different conclusions that point at specific industries, countries, or certain years showing disproportionate impact, likely driven by broader economic events.

Next Step:

The issue with the current dataset is that it is tech focused, making it impossible to draw market wide trends from it. Also, tech companies are going to be focused in majority first world countries, meaning it is also impossible to draw worldwide trends from it. So, given more time and resources, I think the next step would be increasing both the quantity and quality of the data, starting by filling out as many of the nulls as possible by researching missing fields, or by finding supplementary datasets that can fill in what I am missing. I would add the population of the countries those companies come from if applicable along with said countries GDP, the number of hired employees as well as the profits, revenue, and the employee wages of those companies. I would also calculate the number of pre layoffs and post layoffs employees and add that to the dataset, The reason you would add this data is to get a more complete picture of what every tech company looks like before, during, and after they are laying off, and thus allows you to control for more variables and potentially identify more trends. You could see the potential purpose of the layoffs, if the next year they hired back a similar number of employees but the employee wages went down and/or the profits and revenues went up, then that would suggest a wage restructuring and might be a year you give less weight to or outright ignore for certain queries. It also allows you to compare countries that are more similar, comparing the layoffs of America to a country from Africa is probably going to give you some massively misleading results, but if a country like the UK was compared to America, their population and GDP are more comparable and could speak more about a possible trend. Finally, since this current data set is for tech companies as I mentioned in the limitations sections, I would also find another data set that contains other non tech companies, and would compare the tech field with others. By doing so, I could get a better picture of the overall market, and could see if the tech field is unique in its layoffs, on par with other fields, or is actually performing better.

What I Learned:

The most valuable thing I learned throughout this project wasn't any single skill, but learning and actually doing the full pipeline. To see a messy CSV file turn into a clean data set imported onto an SQL Server, and then to see the queries I wrote become a written analysis, all while keeping the question I had chosen at the beginning in mind and using that to guide my actions, that was the most important thing I learned. But during that process I learned a lot of other skills. I utilized the Python library pandas to do an analysis on the data quality and clean the data according to my overall goal. I used the SQL Server extension on VS Code to not only import the clean CSV into my Azure SQL Server, but also to query the newly imported data. On top of picking up

some technical skills, I also learned some conceptual data analysis skills, for example, when cleaning the data, I learned when I drop some columns but keep others, when it's advantageous to keep nulls, fix them, or drop the row entirely. When analyzing the data, I learned how to set thresholds to improve data context, and how to defend those thresholds on analytical grounds. Some of the more unintuitive thought processes I strengthened were how to determine when the incomplete data is worse than no data at all, when maybe changing certain thresholds under certain circumstances is beneficial, and how to not go beyond the scope I set for myself at the very first step.